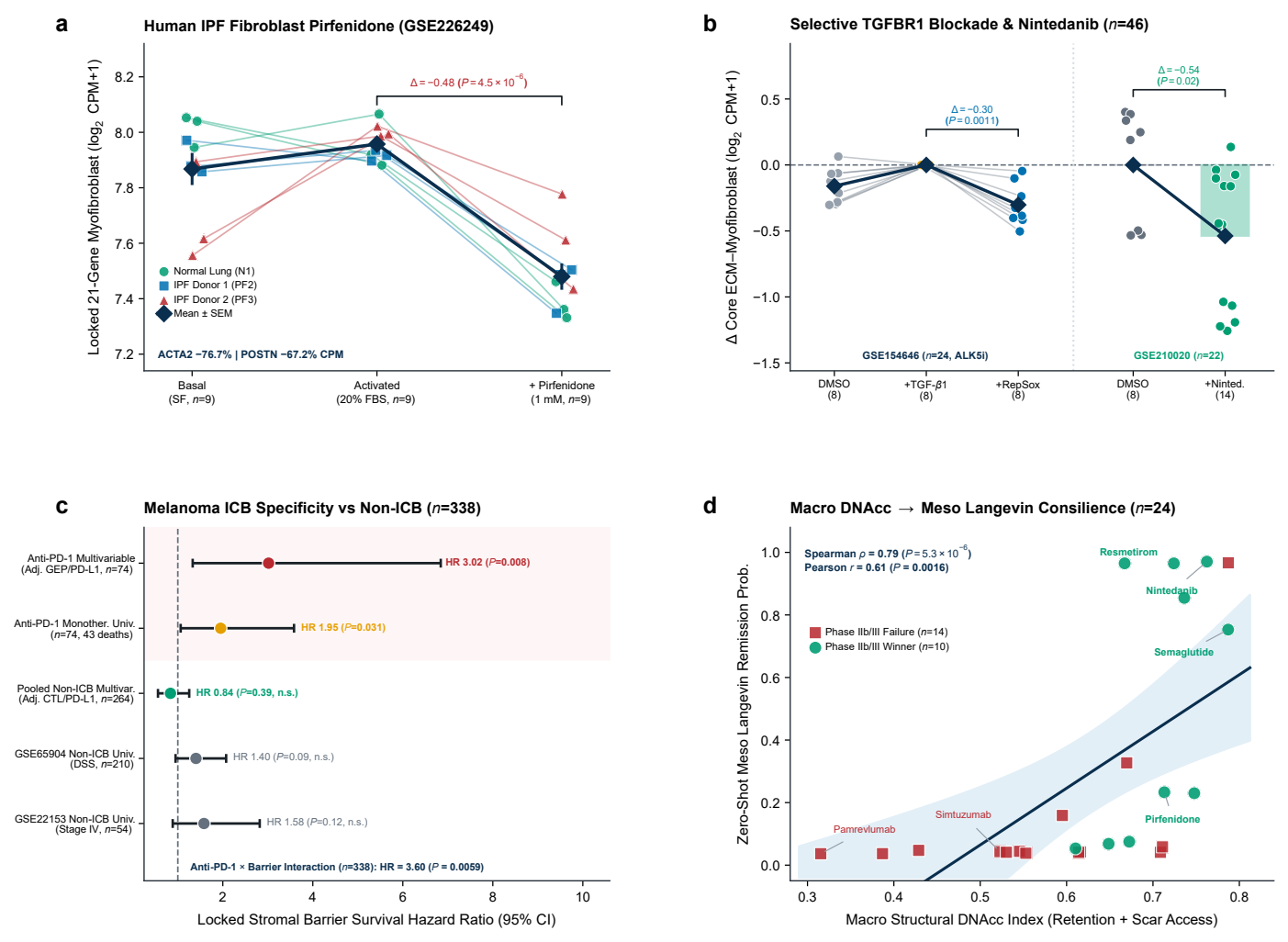

a
Human IPF Fibroblast Pirfenidone (GSE226249)
Locked 21-Gene Myofibroblast (log2 CPM+1)
Δ = −0.48 (P = 4.5 × 10−6)
Normal Lung (N1)
IPF Donor 1 (PF2)
IPF Donor 2 (PF3)
Mean ± SEM
ACTA2 −76.7% | POSTN −67.2% CPM
Basal (SF, n=9)
Activated (20% FBS, n=9)
+ Pirfenidone (1 mM, n=9)
b
Selective TGFBR1 Blockade & Nintedanib (n=46)
Δ Core ECM–Myofibroblast (log2 CPM+1)
Δ = −0.30 (P = 0.0011)
Δ = −0.54 (P = 0.02)
GSE154646 (n=24, ALK5i)
GSE210020 (n=22)
DMSO (8)
+TGF-β1 (8)
+RepSox (8)
DMSO (8)
+Ninted. (14)
c
Melanoma ICB Specificity vs Non-ICB (n=338)
Anti-PD-1 Multivariable (Adj. GEP/PD-L1, n=74)
HR 3.02 (P=0.008)
Anti-PD-1 Monother. Univ. (n=74, 43 deaths)
HR 1.95 (P=0.031)
Pooled Non-ICB Multivar. (Adj. CTL/PD-L1, n=264)
HR 0.84 (P=0.39, n.s.)
GSE65904 Non-ICB Univ. (DSS, n=210)
HR 1.40 (P=0.09, n.s.)
GSE22153 Non-ICB Univ. (Stage IV, n=54)
HR 1.58 (P=0.12, n.s.)
Anti-PD-1 × Barrier Interaction (n=338): HR = 3.60 (P = 0.0059)
Locked Stromal Barrier Survival Hazard Ratio (95% CI)
d
Macro DNAcc → Meso Langevin Consilience (n=24)
Spearman ρ = 0.79 (P = 5.3 × 10−6)
Pearson r = 0.61 (P = 0.0016)
Phase IIb/III Failure (n=14)
Phase IIb/III Winner (n=10)
Resmetirom
Nintedanib
Semaglutide
Pirfenidone
Simtuzumab
Pamrevlumab
Zero-Shot Meso Langevin Remission Prob.
Macro Structural DNAcc Index (Retention + Scar Access)